

Engineering Analysis Report: Bosch HA-M Speed Sensor

Frequency Response, Operating Error Boundaries, and Real-World Velocity Mapping

1. Sensor Operating Profile & Noise Envelopes

The Bosch Motorsport **HA-M** speed sensor relies on an integrated signal conditioning circuit optimized for high-reliability environments. Unlike alternative variants (such as the HA-D 90), the HA-M functions within a highly specific frequency profile, mapping distinct error boundaries and physical limitations.

- **Standard Nominal Envelope (< 4% Error):** Operating from 0 Hz up to $4,200 \text{ Hz}$ (4.2 kHz), the sensor delivers high accuracy, keeping edge-repeatability and signal timing distortions under 4% error.
- **Critical Lag/Clipping Zone (> 4.2 kHz):** Exceeding the upper physical ceiling of 4.2 kHz forces the internal electronics into a soft-clipping state. Instead of just scaling up noise gracefully, the sensor encounters phase lag, signal attenuation, and will ultimately drop pulses entirely.

2. At What Bike Speed Does This Happen?

To evaluate how this 4.2 kHz ceiling maps onto actual vehicle velocities, we model a direct wheel-hub application. The conversion depends strictly on the physical configuration of the tone ring and the tire size. For this analysis, we assume a industry standard **29-inch e-bike wheel** with a total outer diameter (OD) of **740 mm**.

Scenario A: Standard 50-Tooth Tone Ring (Factory ABS Style)

If using a standard 50-tooth ABS-style tone ring directly mounted to the wheel hub, the wheel rotational velocity (RPM) required to reach the $4,200 \text{ Hz}$ threshold is derived as follows:

$$\text{Wheel RPM} = \frac{4200 \text{ Hz} \times 60}{50 \text{ teeth}} = 5,040 \text{ RPM}$$

Converting this rotational velocity to linear ground speed:

$$\text{Speed} = \frac{5,040 \text{ RPM} \times \pi \times 740 \text{ mm} \times 60}{1,000,000} \approx 703 \text{ km/h} \quad (437 \text{ mph})$$

Takeaway: With a standard low-tooth-count tone ring, vehicle velocities will **never** reach or cause this sensor to lag or fail during any realistic bike usage. The simulation model will stay perfectly stable within the safe, high-accuracy (< 4% noise) envelope.

Scenario B: High-Density 120-Tooth Tone Ring (Aftermarket / Development Rig)

For high-fidelity data acquisition platforms, advanced traction control development, or custom aftermarket telemetry setups where a denser 120-tooth target wheel is utilized:

$$\text{Wheel RPM} = \frac{4200 \text{ Hz}}{120 \text{ teeth}} \times 60 = 2,100 \text{ RPM}$$

Converting this higher-resolution profile to linear ground speed:

$$\text{Speed} = \frac{2,100 \text{ RPM} \times \pi \times 740 \text{ mm} \times 60}{1,000,000} \approx 293 \text{ km/h} \quad (182 \text{ mph})$$

Takeaway: On a dense aftermarket 120-tooth target wheel, the sensor maintains stability far past any typical bicycle operating envelope, with the critical boundary layer pushed safely out to track speeds of 293 km/h.

3. Velocity to Frequency Cross-Reference Matrix

The matrix below highlights the behavior of the Bosch HA-M sensor across different combinations of tone ring densities and actual vehicle speeds using the standard 740 mm tire profile.

Tone Ring Profile	Vehicle Speed	Wheel RPM	Output Frequency	Sensor Status & Error Rate
Standard 50-Tooth (Stock)	25 km/h (15.5 mph)	179 RPM	149 Hz	Normal Envelope (< 4% Error)
Standard 50-Tooth (Stock)	60 km/h (37.3 mph)	430 RPM	358 Hz	Normal Envelope (< 4% Error)
Standard 50-Tooth (Stock)	703 km/h (437 mph)	5,040 RPM	4,200 Hz	Absolute Ceiling / Lag Boundary
Aftermarket 120-Tooth (Dense)	25 km/h (15.5 mph)	179 RPM	358 Hz	Normal Envelope (< 4% Error)
Aftermarket 120-Tooth (Dense)	60 km/h (37.3 mph)	430 RPM	860 Hz	Normal Envelope (< 4% Error)
Aftermarket 120-Tooth (Dense)	293 km/h (182 mph)	2,100 RPM	4,200 Hz	Absolute Ceiling / Lag Boundary

4. Simulation Guidance Summary

When programming or parameterizing your simulation frame for the Bosch HA-M sensor, implement a hard threshold at **4,200 Hz**. Any input dynamics tracking below this threshold should be subjected to standard white or pink noise variances bounded strictly under **4%**. If the calculated physical input exceeds 4,200 Hz, the simulation logic must introduce phase-lag attenuation and exponential signal dropouts to reflect the real-world operational failure of the hardware conditioning circuits.